Interacting with External Data Sources

We will examine how Spark SQL interfaces with external components. Specifically, we will discuss how Spark SQL allows you to:

- Use user-defined functions for both Apache Hive and Apache Spark.
- Connect with external data sources such as JDBC and SQL databases, PostgreSQL, MySQL, Tableau, Azure Cosmos DB, and MS SQL Server.
- Work with simple and complex types, higher-order functions, and common relational operators.

Querying with Spark SQL Shell and Beeline

- Spark SQL provides a command line interface called the Spark SQL shell. It is similar to the Spark shell, but it is designed for running SQL queries. It can be started by running

```
spark-sql
```

- Now we can use it interactuvely to run SQL queries against data stored in Hive, Parquet, JSON, and other formats. Let's create a simple table and insert some data into it.

```
spark-sql> CREATE TABLE people (name STRING, age INT);
spark-sql> INSERT INTO people VALUES ('Alice', 30);
spark-sql> INSERT INTO people VALUES ('Michael', NULL);
spark-sql> INSERT INTO people VALUES ('Samantha', 39);
```

- Now we can query the table using SQL queries.

```
spark-sql> SHOW TABLES;
spark-sql> SELECT * FROM people;
spark-sql> SELECT name FROM people WHERE age IS NULL;
spark-sql> quit;
```

DataFrameReader Parameters

Method	Arguments	Description
`format()`	`"parquet", "csv", "txt", "json", "jdbc", "orc", "avro"` etc	Default is parquet or as specified in the config for `spark.sql.sources.default`.
`option()`	`("mode", {PERMISSIVE | FAILFAST | DROPMALFORMED )`	default is `PERMISSIVE`
	`("inferSchema", {true | false})`	`mode` and `inferSchema` is for JSON and CSV sources
	`("path", "path_file_data_source")`	For JSON and CSV formats
`schema()`	DDL String or StructType	
`load()`	`"path/to/data/source"`	optional if already specified above

DataFrameReader Parameters (cont.)

```
# Use Parquet
file = """flights/data/summary-data/parquet/2010-summary.parquet"""
df = spark.read.format("parquet").load(file)

# Use Parquet; you can omit format("parquet") if you wish as it's
    the default
df2 = spark.read.load(file)

# Use CSV
df3 = (spark.read.format("csv")
       .option("inferSchema", "true")
       .option("header", "true")
       .option("mode", "PERMISSIVE")
       .load("flights/data/summary-data/csv/*"))

# Use JSON
df4 = (spark.read.format("json")
       .load("flights/data/summary-data/json/*"))
```

DataFrameWriter

- `DataFrameWriter` is the core construct for writing data from a DataFrame to external source a DataFrame. Recommended formats:

```
DataFrameWriter.format(args)
                .option("key", "value")
                .bucketBy(args)
                .partitionBy(args)
                .save(path)
```

```
DataFrameWriter.format(args).option("key", "value")
               .sortBy(args).saveAsTable(table)
```

- We can access a `DataFrameWriter` directly through a `DataFrame` instance by using either `DataFrame.write` or `DataFrame.writeStream`.

DataFrameWriter Parameters

Method	Arguments
`format()`	`"parquet", "csv", "txt", "json", "jdbc", "orc", "avro"` etc
`option()`	`("mode", {append | overwrite | ignore | error or errorifexists} )`
	`("mode", {SaveMode.Overwrite | SaveMode.Append | SaveMode.Ignore | SaveMode.ErrorIfExists})`
	`("inferSchema", {true | false})`
	`("path", "path_to_write_to")`
`bucketBy()`	#buckets and names of columns to bucket by
`save()`	`"path/to/data/source"`
`saveAsTable()`	`"table_name"`

Overview of Data Source Types

- Parquet
- JSON
- CSV
- Avro: Used by Apache Kafka for serialization. Offers direct mapping to JSON, speed and efficiency, and bindings for many programming languages.
- ORC (Optimized Row Columnar) file format provides a highly efficient way to store Hive data.
- Image data
- Arbitrary binary data: The DataFrame has the following attributes: path, modificationTime, length, and content. Writing isn't supported.

Parquet

- Parquet is a columnar storage format that is available to any project in the Hadoop ecosystem, regardless of the choice of data processing framework, data model, or programming language.
- Parquet is a popular choice for data storage and processing because it is efficient, uses columnar storage, and employs a fast compression algorithm.
- Parquet is the default data source format for Spark.

External Data Sources

- Popular tools like Tableau and PowerBI can connect simply to spark engine using the Spark URL (like we do for connecting to the cluster).
- We can also easily connect to SQL data bases using the JDBC connector and appropriate jar file for PostgreSQL, MySQL, SQLite, Azure Cosmos DB, MS SQL Server.
- Connect to distributed databases, data lakes such as Cassandra, MongoDB, Snowflake, etc. For example, see: https://github.com/datastax/spark-cassandra-connector